

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

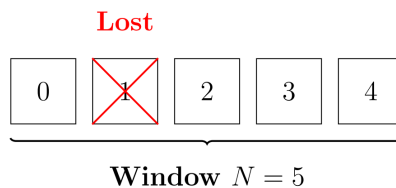
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Section:

Q1. Carefully read the Question before attempting: [5 marks]

Consider a pipelined protocol over a link with a very high Bandwidth-Delay Product. The sender sends packets 0, 1, 2, 3, 4. Packet 1 is lost. All other packets and ACKs arrive correctly. Timeout occurs after Packet 4 is sent. Window Size $N = 5$. Sequence Numbers: 0, 1, 2, ...

- If using **Go-Back-N**, mathematically calculate the total number of packet transmissions (initial + retransmissions) required to successfully deliver packets 0 through 4. If using **Selective Repeat**, calculate the same.
- Will **Selective Repeat** operate correctly if $N = 5$ and $k = 3$? Consider the SR window size constraint, create a scenario where it fails and explain it (why)?



Solution:

(a)

- Go-Back-N:** 1. Sender sends 0, 1, 2, 3, 4. (5 transmissions). 2. Receiver gets 0 (ACKs 0). Receiver expects 1, gets 2, 3, 4 (discards them, re-ACKs 0). 3. Sender Timeout for Packet 1. 4. GBN retransmits from base: Sends 1, 2, 3, 4. (4 transmissions). Total transmissions = $5 + 4 = 9$.
- Selective Repeat:** 1. Sender sends 0, 1, 2, 3, 4. (5 transmissions). 2. Receiver gets 0 (ACKs 0). Receiver misses 1. Receiver gets 2, 3, 4 (Buffers them, ACKs 2, 3, 4 selectively). 3. Sender Timeout for Packet 1. 4. Sender retransmits ONLY 1. (1 transmission). Total transmissions = $5 + 1 = 6$.

(b)

SR Protocol Constraint: Window Size N must satisfy $N \leq \frac{1}{2} \times 2^k$.

Here $2^k = 8$. Max N should be 4.

Given $N = 5$. Problem:

- Sender sends 0,1,2,3,4.
- Rx receives all, sends ACKs, moves window to [5,6,7,0,1].
- All ACKs lost.
- Sender times out, resends Packet 0.
- Receiver sees Packet 0. Is this a retransmission of old 0 or new 0 (in window [5,6,7,0,1])? It cannot distinguish. Protocol fails.

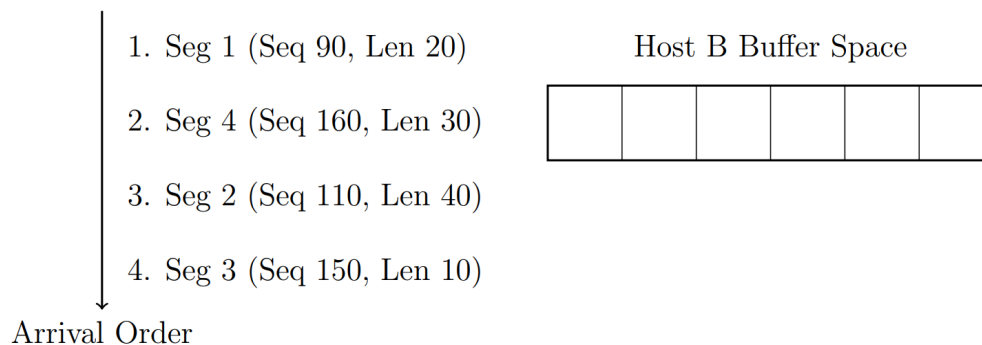
Q2. Carefully read the Question before attempting:**[10 marks]**

Host A sends 4 segments to Host B over a TCP connection.

- Segment 1: Seq = 90, Length = 20 bytes.
- Segment 2: Seq = 110, Length = 40 bytes.
- Segment 3: Seq = 150, Length = 10 bytes.
- Segment 4: Seq = 160, Length = 30 bytes.

Assume the initial Sequence Number is 90. Host B receives them in this order: Segment 1, Segment 4, Segment 2, Segment 3.

- (a) List the ACK number Host B sends immediately after receiving EACH segment in that specific order.
- (b) Draw the receive buffer state after Segment 4 arrives but before Segment 2 arrives.

**Solution:****(a) ACK Generation:**

1. Rx Seg 1 (90-109): Data is in order. Next expected is 110. **ACK = 110.**
2. Rx Seg 4 (160-189): Gap detected (missing 110-159). Buffer Seg 4. Send duplicate ACK. **ACK = 110.**
3. Rx Seg 2 (110-149): Gap partially filled. Still missing 150. Buffer Seg 2. Send duplicate ACK. **ACK = 150.**
4. Rx Seg 3 (150-159): Gap filled. We have 90 through 189 continuous bytes. Next expected is 190. **ACK = 190.**

(b) Receive Buffer State (After Seg 4, Before Seg 2):

State Description: Segment 1 is delivered/acknowledged. Segment 4 is buffered (out of order). There is a "Hole" where Segment 2 and 3 should be.

